Evaluating the Impact of Image Augmentations on Object Detection Models

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Abstract

Data augmentation is widely used in object detection to improve robustness and generalization, yet augmentation strategies are often chosen by convention rather than through systematic evaluation. In this work, we present a controlled study of individual image augmentations and their effects on object detection performance. Using YOLOv11n as the detection model, we evaluate a range of geometric, photometric, and occlusion-based transformations under a strictly controlled experimental setup, where only a single augmentation and its application probability are varied per run. Experiments are conducted on two datasets with distinct characteristics: KITTI, which consists of structured street scenes with a limited number of classes, and COCO-mini, a diverse multi-class subset of the COCO benchmark. Performance is measured using mAP@0.5:0.95 on unaugmented validation sets. Across 74 training runs, we find that augmentation effectiveness is highly dataset-dependent: some transformations yield consistent improvements, while others degrade performance when applied aggressively. We further observe that lower application probabilities often outperform always-on augmentation, highlighting the importance of augmentation strength. Overall, our findings show that augmentations are not universally beneficial and underscore the need for dataset-aware, empirically validated augmentation strategies in object detection pipelines.

1 Introduction

In modern computer vision, data augmentation has become a key technique for improving model robustness and generalization. By applying controlled image transformations such as flipping, rotation, scaling, cropping, and color adjustments, augmentation increases the effective size and diversity of training data, helping models handle variations common in real-world environments. However, the effectiveness of different augmentation strategies can vary significantly across datasets and tasks.

This project aims to provide insights into which image transformations most improve detection accuracy and training stability in practical object detection settings. Rather than evaluating complex augmentation pipelines, we focus on controlled, single-augmentation experiments to better isolate and understand the contribution of individual transformations.

1.1 Project Type

Self-defined research.

2 Motivation

The motivation for this project arises from the need to identify efficient and effective augmentation methods that can improve the performance of object detection models across different datasets.

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By evaluating a range of augmentation techniques under controlled conditions, we aim to develop practical insights that can guide the design of data-efficient training pipelines.

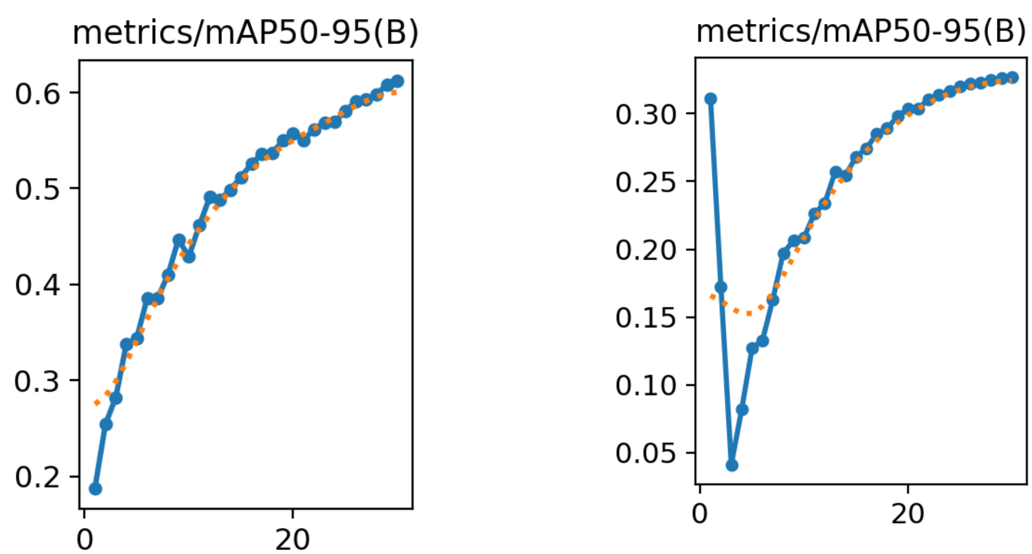
Understanding how augmentation effectiveness varies across datasets is particularly important for real-world applications, where data availability and domain characteristics can differ significantly. The findings from this study serve as a foundation for future work on augmentation selection and optimization, supporting the development of more reliable and robust computer vision models.

3 Approach

3.1 Baseline Approach

The baseline approach involves training YOLOv11n object detection models on the KITTI Object Detection dataset [2] and the COCO-mini dataset [6] without applying any data augmentations, including model-internal augmentation mechanisms. Each dataset uses its own optimized hyper-parameters (e.g., learning rate, batch size, and number of epochs). However, within each dataset, the baseline and subsequent augmentation experiments share the same configuration to ensure fair comparison.

YOLOv11 [3] is a modern real-time object detection model in the YOLO (You Only Look Once) family that emphasizes a balance between accuracy, speed, and training stability. It follows a single-stage detection paradigm, predicting bounding boxes and class probabilities directly from full images in a single forward pass. Compared to earlier YOLO versions, YOLOv11 incorporates architectural and training refinements that improve localization and mAP performance while maintaining efficient inference. Its support for transfer learning from pretrained COCO weights enables fast convergence and strong performance when fine-tuned on smaller or domain-specific datasets.



(a) KITTI

(b) COCO-mini

Figure 1: Baseline (no-augmentation) training curves showing mAP@0.5:0.95 versus epoch.

3.2 Advanced Approach

To evaluate the effects of data augmentation, we apply a set of geometric, photometric, and occlusion-based transformations to the KITTI and COCO-mini training datasets using the Albumentations library [1]. Augmentations are applied in isolation, with exactly one transformation enabled per experiment.

Each augmentation is applied with a fixed probability, allowing us to analyze how augmentation strength influences training behavior and generalization. All other training settings are held con-

Table 1: Augmentation operators evaluated in this study, their category, dataset applicability, and fixed strength constraints. Checkmarks indicate that the augmentation was applied for the corresponding dataset.

Augmentation	Type	KITTI	COCO-mini	Strength / Constraints
Mirror	Geometric	✓	✓	Horizontal flip
Rotate	Geometric	✓	✓	$\pm 30°$ rotation, border reflect
Zoom	Geometric	✓	✓	Affine scale 0.8–1.2
Crop	Geometric	✓	✓	BBox-safe crop to 640×640
Brightness	Photometric	✓	✓	Brightness limit 0.2
Contrast	Photometric	✓	✓	Contrast limit 0.2
Sharpness	Photometric	✓	✓	Alpha 0.1–0.3, lightness 0.7–1.3
Blur	Photometric	✓	✓	Motion / Gaussian / Median blur
Dropout	Occlusion	✓	✓	1–8 holes, 5–20% image size
Affine (small)	Geometric	✓	–	Scale 0.98–1.02, translate/rotate/shear $\pm 2\%$
HSV	Photometric	✓	–	Hue ± 5, Sat ± 15, Val ± 10
Gamma	Photometric	✓	–	Gamma 80–120
CLAHE	Photometric	✓	–	Clip limit 1–3, grid 8×8
Gaussian noise	Photometric	✓	–	Variance 5–25
Motion blur (small)	Photometric	✓	–	Blur limit = 3

BBox filtering: min_visibility = 0.3 (boxes dropped if overly occluded after geometric transforms).
Image size: imgsz = 640 (YOLO input resolution).
Augmentation probability: $p \in \{1.0, 0.5, 0.25\}$.

stant between baseline and augmented runs. This controlled design enables direct attribution of performance changes to individual augmentations.

The entire augmentation–training sweep is fully automated. Configuration generation, training execution, metric extraction, and result aggregation are handled programmatically. While this pipeline is effective for exploring individual augmentations, the augmentation search space grows exponentially as more transformations, probabilities, and strength parameters are considered. Even under the restricted single-augmentation setting, this study required 74 full training runs.

3.3 Pipeline Diagram

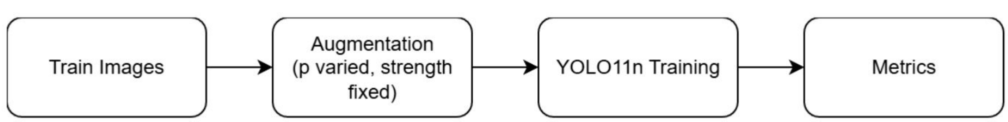


Figure 2: Automated sweep–augment–train–evaluate pipeline used in all experiments.

3.4 Libraries and Computational Resources

Albumentations [1] is used to implement geometric, photometric, and occlusion-based augmentations with correct bounding-box transformations. Ultralytics YOLO [3] is used to train YOLOv11n and compute validation metrics. All experiments are conducted using single-GPU training, and overall compute cost is dominated by repeated full training runs across the sweep.

4 Experiments

4.1 Data

4.1.1 KITTI

The KITTI dataset[2] consists of 7,824 labeled images captured from a vehicle-mounted camera in urban driving environments. Images depict structured street scenes with relatively consistent viewpoints and lighting conditions. The dataset contains 8 object classes relevant to autonomous driving scenarios. For this study, the dataset is split into 5,984 training images and 748 validation images (approximately an 80/20 split), and all evaluation is performed on the validation subset.

Figure 3: Example image from the KITTI dataset.

4.1.2 COCO-mini

COCO-mini[6] is a reduced subset of the COCO dataset[4] designed to retain the diversity of the original benchmark while remaining computationally manageable. It contains approximately 30,000 images spanning a wide range of scenes, object categories, and visual conditions. The dataset includes 80 object classes and is split into 25,000 training images and 5,000 validation images. Compared to KITTI, COCO-mini exhibits significantly greater visual and semantic diversity, making it a more challenging multi-class detection benchmark.

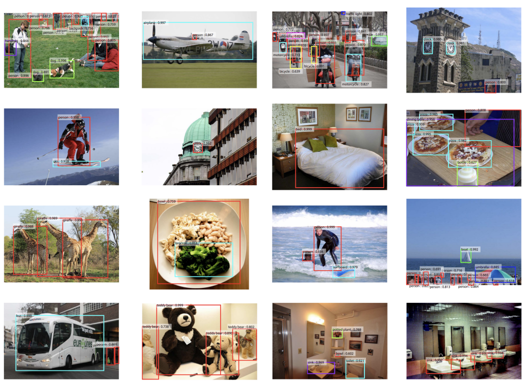


Figure 4: Example images from the COCO-mini dataset.

4.2 Metric

We report mAP@0.5:0.95 on the unaugmented validation set. This COCO-style metric evaluates detection performance across multiple IoU thresholds and captures both localization accuracy and classification quality [4].

4.3 Augmentation Set and Constraints

All experiments isolate a single augmentation operator implemented using Albumentations [1]. Bounding boxes are transformed jointly with images, and boxes with insufficient visibility are removed after geometric transformations. A minimum visibility threshold of 0.3 is used. All images are resized to 640×640 prior to training. Augmentation probability is swept over $p \in \{1.0, 0.5, 0.25\}$.

Figure 5: Visual examples of augmentation operators evaluated in this study.

5 Evaluation and Validation

Baseline training curves (Figure 1) confirm stable convergence on both datasets and provide context for interpreting augmentation-induced gains.

5.1 Single-Augmentation Results

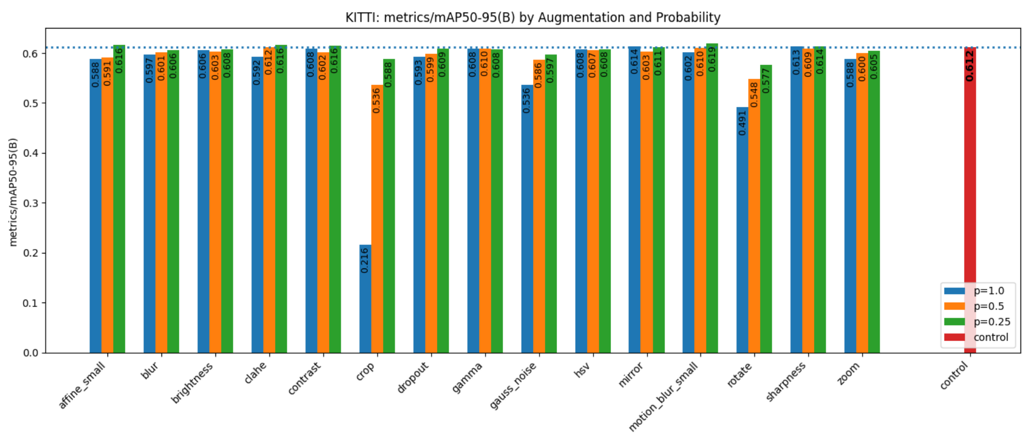


Figure 6: KITTI validation mAP@0.5:0.95 by augmentation and probability.

On KITTI, several augmentations improve performance when applied with lower probability, while always-on augmentation often degrades results. This suggests that mild regularization is beneficial for structured datasets [2].

On COCO-mini, augmentations are largely neutral or slightly harmful, consistent with the model already being pretrained on COCO [4] and the dataset itself exhibiting high diversity.

6 Conclusion and Discussion

6.1 Insights

Augmentation effectiveness is strongly dataset-dependent. Lower augmentation probability often outperforms always-on augmentation, particularly on structured datasets such as KITTI[2]. Pretraining on COCO[4] reduces the marginal benefit of augmentation on COCO-like datasets, making gains more difficult to achieve. Automated single-augmentation sweeps are a practical diagnostic tool for identifying safe and harmful augmentations before exploring more complex combinations.

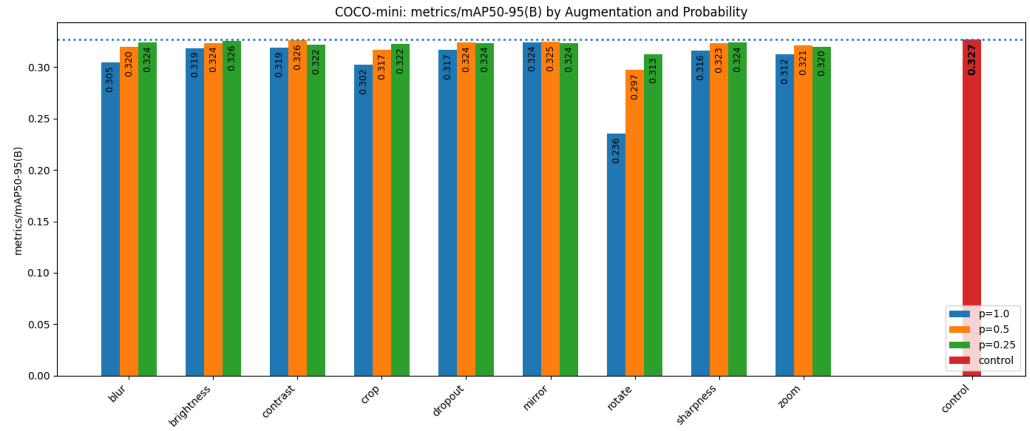


Figure 7: COCO-mini validation mAP@0.5:0.95 by augmentation and probability.

6.2 Workload

Cameron Tucker implemented the augmentation pipeline, automated the experiment sweeps, generated visualizations, and performed the primary analysis and writing. Zhiyuan Chen prepared datasets, configured YOLO training, validated experiments, and assisted with result organization and report refinement.

6.3 Key challenges

A major challenge was managing computational cost while maintaining experimental rigor. Our original project plan proposed using RF-DETR[5]; however, preliminary experiments revealed that RF-DETR training time per run was prohibitively expensive for large-scale augmentation sweeps. Because this study requires dozens of full training runs to isolate individual augmentation effects, RF-DETR would have severely limited the number of configurations we could evaluate. We therefore transitioned to YOLOv11n, which provides significantly faster training and more stable convergence while remaining a strong, widely used object detection baseline. This change enabled us to conduct 74 controlled experiments within available compute constraints while preserving fair comparisons across augmentations.

A second challenge was dataset scale. Training on the full COCO dataset would have been infeasible given the need to repeat training across many augmentation settings. To address this, we used COCO-mini[6], a reduced yet diverse subset of COCO that preserves multi-class complexity while remaining computationally tractable. This allowed us to study augmentation behavior in a challenging, high-diversity setting without sacrificing experimental breadth or consistency.

6.4 Future Work

This study evaluates only single augmentations applied in isolation in order to clearly attribute performance changes to individual transformations. In practice, detection pipelines often apply multiple augmentations sequentially, and interactions between augmentations may produce non-linear effects. Exploring such combinations remains an important direction for future work, but the search space grows exponentially with the number of augmentations, probabilities, and strength parameters considered.

Future extensions of this work could explore multi-augmentation pipelines guided by the best-performing single augmentations identified here, expand the sweep to include additional probability and strength values, and evaluate augmentation behavior across more datasets, domains, and model architectures. The fully automated sweep–train pipeline developed in this project provides a strong foundation for scalable exploration of these directions.

References

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